Graph Contrastive Learning and Multigraph Attention Fusion Network for Hyperspectral Image Change Detection

Jie Yang, Jiangtao Peng, Bing Yang, and Weiwei Sun

***Abstract*—In recent years, graph neural networks (GNNs) have been increasingly applied to the hyperspectral image change detection (HSI-CD). However, existing methods propagate features by weighting nodes and edges, which may weaken the original node characteristics and spectral differences. Moreover, current methods typically treat bitemporal features in isolation and lack intertemporal semantic modeling. To address the above issues, this article proposes a graph contrastive learning and multigraph attention fusion network (GCLMA), which leverages the contrastive learning to extract discriminative features and enhances semantic relevance through graph attention-based interactions across multiple graphs. The proposed GCLMA method mainly consists of four modules, i.e., spatial–spectral graph construction (SSGC), contrastive feature learning ResGCN (CFLR), differential feature enhancement (DFE), and multigraph interactive fusion (MGIF). The SSGC module is first designed to construct a well-defined graph structure by adaptively integrating the spatial proximity and spectral similarity. Then, the CFLR module utilizes the contrastive learning to extract discriminative feature representations that are sensitive to subtle changes. Next, the DFE module amplifies significant spectral variations to emphasize critical change regions. Finally, the MGIF module employs the graph attention mechanism to interactively fuse enhanced difference graph features with bitemporal graph features, thereby facilitating the semantic modeling. Experimental results on three HSI-CD datasets show that the proposed GCLMA outperforms most existing state-of-the-art methods. The source code of the proposed method will be released at: https://github.com/YYYYYJJJJJJ/GCLMA**

***Index Terms*—Change detection (CD), contrastive learning, graph attention network, hyperspectral image (HSI).**

I. Introduction

THE change detection (CD) aims to identify land cover changes by analyzing a pair of remote sensing images, known as bitemporal images, captured over the same area at different times. As a type of remote sensing image, hyperspectral images (HSIs) typically contain hundreds or even thousands of spectral bands. This remarkable characteristic enables the HSI-CD task to identify changes more accurately, as each material possesses a unique and corresponding spectral fingerprint [1], [2], [3]. During the past few years, HSI-CD has been widely used in various fields, including meteorological monitoring [4], forestry detection [5], disaster detection [6], and urban expansion [7].

Traditional HSI-CD approaches can be divided into three primary categories: algebra-based methods, transformation-based methods, and classification-based methods. Algebra-based methods usually rely on similarity thresholds to identify change regions, with common methods such as regression, difference, ratio, and change vector analysis (CVA) [8]. The CVA uses the Euclidean distance to measure change between bitemporal images. Due to the simplicity and effectiveness, many CVA variants are proposed, such as robust CVA (RCVA) [9] and compressed CVA (C^2VA) [10]. RCVA improves CVA by considering pixel neighborhoods to mitigate radiometric and geometric distortions. The C^2VA reduces the information loss by using a compressed 2-D representation. Algebraic methods are heavily dependent on thresholds and are sensitive to noise. Transformation-based methods transform the image from the spatial domain to the feature domain and compare features over time to identify the changed regions. The most common method is the principal component analysis (PCA) [11]. It extracts orthogonal feature vectors from nonoverlapping blocks of the different map to form a feature space and then classifies pixels through clustering. The slow feature analysis [12] method extracted unchanged components by leveraging the slow changes in pixels between bitemporal images. In addition, there are some other transform-based methods, such as iteratively reweighted multivariate alteration detection (IR-MAD) [13] and deep belief networks (DBNs) [14]. These methods can suppress noise but with some loss of spectral information. The classification-based approach treats the CD task as a classification task. Commonly used methods include support vector machines [15], extreme learning machines [16], and maximum likelihood classifiers [17]. Traditional methods depend on hand-crafted features that are typically dataset specific. This often leads to reduced discriminative performance.

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Deep learning's powerful feature extraction capabilities have advanced the HSI-CD task. Particularly, convolutional neural networks (CNNs) can extract hierarchical local spatial dependencies and have been widely used in HSI-CD. For example, Luo et al. [18] proposed a multiscale diff-changed feature fusion network (MSDFFN), which utilized an encoder–decoder subnetwork for temporal feature extraction and convolutional layers for CD at multiple scales. Wang et al. [19] proposed an end-to-end 2-D network that introduced a hybrid affinity matrix to integrate subpixel representations and used 2-D CNN to extract discriminative features. Li et al. [20] used a 2-D spatial–spectral self-attention module to extract different information by considering the correlation between adjacent pixels. Yang et al. [21] introduced a deep multiscale pyramid network to enhance spatial–spectral residual attention for HSI-CD. Recently, Zhou et al. [22] designed a low-rank and sparse estimation module along with a CD network to extract change information. Jiang et al. [23] proposed a Siamese CNN network that integrated adaptive spectral–spatial attention (SA) and center-focused attention. Chen et al. [24] proposed a temporal difference-guided network that hierarchically extracted spectral features and fused multilevel spatial differences via convolution. Although CNN has advantages in extracting spatial local features, it is difficult to capture global contextual information effectively due to its local receptive field. Different from the CNN model, transformer-based networks can capture global dependencies using the self-attention mechanism. Wang et al. [25] proposed a spectral–spatial–temporal transformer to extract relevant features. Sun et al. [26] proposed a spatial–spectral transformer CD network (STCD-Former) that generated spectral markers from different spectral sequences to guide the learning of change rules. Zhang et al. [27] proposed a transformer-based correlation representation module that captured the spectral dependency using disparity and dot product operations. Wang et al. [28] developed a domain alignment transformer (DA-Former), which aligned bitemporal HSI features across domains. Zhang et al. [29] designed a Siamese U-shaped transformer for CD. Wu et al. [30] employed a cross-attention mechanism to extract and enhance interaction features from bitemporal images. Seydi et al. [31] proposed a dual-branch Siamese network that combined the CNN and transformer, where the transformer-based spectral branch leveraged self-attention to model complex spectral dependencies.

In recent years, graph convolutional networks (GCNs) have also received a lot of attention because of their ability to flexibly model spatial dependencies [32], [33], [34]. For instance, Yang et al. [35] proposed a Laplacian sharpening-based convolutional module to counteract the reduction of differences caused by Laplacian smoothing in graph convolutions. Qu et al. [36] proposed a dual-branch differential amplified graph convolution network (D^2AGCN), which used a two-branch structure and injected differential information into each graph convolutional layer. Yang et al. [37] proposed a fine–coarse–fine progressive graph framework that generated hierarchical segmentations from fine to coarse and integrated features progressively to improve pixelwise CD. Zhao et al. [38] proposed a graph transformer-guided multitemporal information united framework for HSI-CD (GTransCD), which used a graph transformer to capture differential features and a gating unit for information fusion. Wang et al. [39] proposed a two-branch network that extracted pixel-level and superpixel-level features and fused them using graph attention. Yang et al. [40] proposed a graph network that fused augmentation and differential information and reconstructed multilevel features using attention modules. Qu et al. [41] proposed a cyclic consistency constrained multiview graph matching network that aligned heterogeneous images in multiple domains to preserve spatial–spectral information and enforce the consistency across time.

Despite the strong performance of GCN-based methods in the HSI-CD task, they still face challenges in learning discriminative feature representations. GCNs update node features by aggregating information about the node itself and its neighbors to capture local and global dependencies. However, the weighted summation mechanism used in this process may lead to overmixing of information. As the network depth increases, the feature propagation may weaken the original node features and spectral differences between nodes, leading to blurring of the representations and thus making it difficult to retain discriminative feature representations. In addition, existing methods typically focus on the extraction of bitemporal features while neglecting intertemporal semantic modeling. Here, intertemporal refers to the correlations, dependencies, or interactions between features of bitemporal HSIs, while semantic refers to the high-level representations derived from both spectral and spatial features that reflect class-level or object-level information. Most approaches adopt simple differencing or concatenation operations to process bitemporal features for prediction. Such operations lack semantic modeling between bitemporal features, potentially overlooking intertemporal correlations.

To address the above problems, we propose a graph contrastive learning and multigraph attention fusion network (GCLMA) for HSI-CD. The proposed GCLMA method employs the contrastive learning strategy to extract discriminative features and enhances semantic relevance through graph attention-based interactions across multiple graphs. It mainly consists of four modules, i.e., spatial–spectral graph construction (SSGC), contrastive feature learning ResGCN (CFLR), differential feature enhancement (DFE), and multigraph interactive fusion (MGIF). The main contributions are as follows.

1) The SSGC is designed to adaptively integrate spatial proximity and spectral similarity, forming a well-structured graph that enables the network to capture both local spatial texture and spectrally correlated information.
2) A novel CFLR is proposed to integrate the contrastive learning to reinforce interclass separability for changed features and intraclass consistency for unchanged ones. It improves sensitivity to subtle changes, enhancing feature distinctiveness.
3) The MGIF is presented to utilize graph attention to facilitate the interaction among graph features and to enhance difference representations, which can

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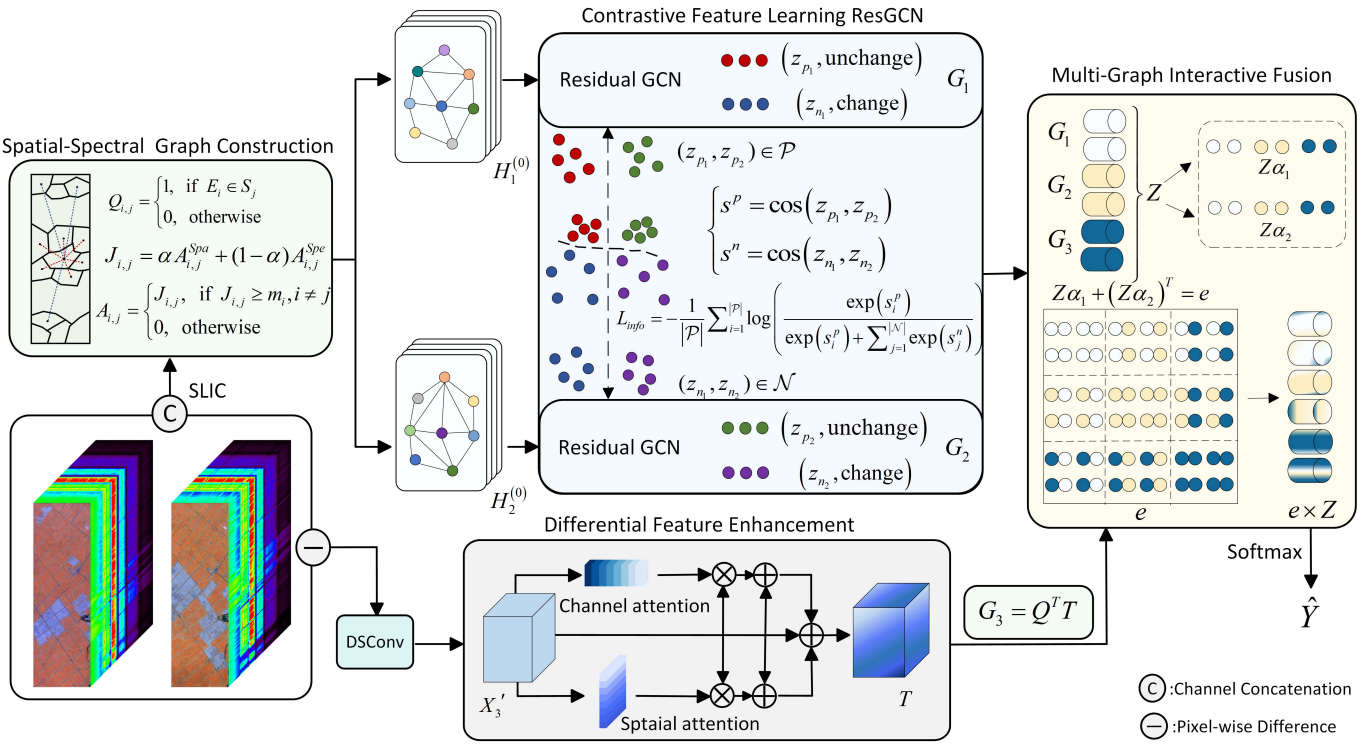


Fig. 1. Framework of the proposed GCLMA. The SSGC module constructs a spatial–spectral graph. The CFLR module obtains discriminative features by introducing contrastive learning. Meanwhile, the DFE module gets the enhanced difference features. Finally, the discriminative features and enhanced difference features are interactively fused in the MGIF module.

capture intertemporal semantics and enrich feature representations.

The remainder of this article can be divided into three sections. Section II describes the proposed network. In Section III, the experimental results and analysis are presented. Section IV concludes this article.

II. Proposed Method

The framework of the proposed network is illustrated in Fig. 1. The process consists of two main parts: graph feature extraction and differential information injection. In the first part, the SSGC module constructs spatially neighboring and spectrally highly similar graph structures, and then, the CFLR module utilizes the features obtained from the final graph convolution layer and enhances the discriminative representation of features by contrastive learning. In the second part, the DFE module obtains enhanced differential features, which together with the graph features produced by the CFLR module, are input to the MGIF module to produce the interaction representation.

A. SSGC Module

Traditional graph structures are constructed on the basis of spatial adjacency, neglecting the use of spectral information. Here, we would like to construct a graph structure that helps the network learn from nodes with high spatial proximity and spectral similarity.

We define the bitemporal HSIs as $X_1 \in \mathbb{R}^{H\times W\times B}$ and $X_2 \in \mathbb{R}^{H\times W\times B}$, and $Y \in \mathbb{R}^{H\times W}$ is the corresponding ground truth, where H, W, and B denote the height, width, and spectral dimension of the HSIs, respectively. The bitemporal HSIs are then cascaded and segmented using the simple linear iterative clustering (SLIC) algorithm [42] to obtain a unified association matrix. Specifically, the bitemporal HSIs are cascaded as $E = X_1 \parallel X_2$, and its pixel set is represented as $E = \{E_i\}_{i=1}^{N}$, where $N = HW$ is the number of total pixels. After superpixel segmentation, the superpixel set $S = \{S_j\}_{j=1}^{K}$ can be obtained, where K denotes the number of superpixels. Based on the correspondence between the set of pixels and the set of superpixels, we can obtain an association matrix $\mathbf{Q} \in \mathbb{R}^{N\times K}$, which is denoted as follows:

$$Q_{i,j} = \begin{cases} 1, & \text{if } E_i \in S_j \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

where $i \in \{1, 2, \ldots, N\}$ and $j \in \{1, 2, \ldots, K\}$. $Q_{i,j}$ denotes the correspondence between pixel E_i and superpixel S_j. Based on the association matrix Q, we transform pixel-level features into graph node features by averaging all pixels within each superpixel set

$$S_j = \frac{1}{K_j}\sum_{i=1}^{K_j} x_j^i \tag{2}$$

where x_j^i denotes the ith pixel belonging to the jth superpixel and K_j is the number of pixels contained in the jth superpixel.

The cosine similarity is used to measure the spectral similarity between graph nodes as follows:

$$\theta_{i,j} = \arccos\left(\frac{S_i S_j^{\mathrm{T}}}{\|S_i\|\,\|S_j\|}\right) \tag{3}$$

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$$A_{i,j}^{\text{Spe}} = 1 - \frac{\theta_{i,j}}{\pi} \tag{4}$$

where $\theta_{i,j}$ and $A_{i,j}^{\text{Spe}}$ denote the spectral vector angle and spectral weight relationship between graph nodes S_i and S_j, respectively.

Then, we compute the spatial position distance between graph nodes. Denote each pixel coordinate as (x_i, y_j), by averaging the pixel coordinates within a superpixel set, we can obtain a superpixel coordinate as follows:

$$c_\ell = (\bar{x}_\ell, \bar{y}_\ell) \tag{5}$$

where c_ℓ denotes the spatial location coordinate of the ℓth superpixel S_ℓ. $\bar{x}_\ell = (1/|K_\ell|)\sum_{(x_i,y_j)\in S_\ell} x_i$, $\bar{y}_\ell = (1/|K_\ell|)\sum_{(x_i,y_j)\in S_l} y_i$ are the mean value of the x-coordinate and y-coordinate of the superpixel S_ℓ. The spatial distance between graph nodes S_i and S_j can be calculated as follows:

$$O_{i,j} = \left\| c_i - c_j \right\|_2. \tag{6}$$

The obtained distance matrix O is normalized and then used to generate the spatial proximity matrix as follows:

$$A_{i,j}^{\text{Spa}} = 1 - O_{i,j}. \tag{7}$$

Next, a dense spatial–spectral similarity matrix that incorporates both spectral similarity and spatial proximity can be constructed as follows:

$$J_{i,j} = \alpha A_{i,j}^{\text{Spa}} + (1-\alpha) A_{i,j}^{\text{Spe}} \tag{8}$$

where $\alpha \in [0, 1]$. By adjusting α, the fusion ratio of the two types of information can be controlled.

To further reduce the information redundancy, we adaptively select neighbor nodes based on the characteristics of the data itself. Specifically, the row mean of J is denoted by $m_i = (1/K)\sum_{j=1}^{K} J_{ij}$. If $J_{i,j} < m_i$ or $i = j$, then the value at that position is set to 0. This can be formulated as follows:

$$A_{i,j} = \begin{cases} J_{i,j}, & \text{if } J_{i,j} \geq m_i, i \neq j \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

As a result, we can obtain an adaptive spatial–spectral adjacency matrix, which enhances the connections between graph nodes and facilitates the propagation of information during feature extraction.

B. CFLR Module

Here, the contrastive learning is used to enhance the discriminative ability of node representations. It encourages the model to produce representations that effectively distinguish changed and unchanged features. In addition, residual connections are introduced in each layer of GCN to maintain the effective transmission of node features.

Denote the graph node matrix as H, the graph convolutional process can be represented as follows:

$$H_i^{(\ell)} = \sigma\left(\hat{D}^{-\frac{1}{2}}\tilde{A}\hat{D}^{-\frac{1}{2}} H_i^{(\ell-1)} W_1^{(\ell-1)} + H_i^{(\ell-1)} W_2^{(\ell-1)}\right) \tag{10}$$

where $W_1^{(\ell-1)}$ and $W_2^{(\ell-1)}$ are the learnable matrices of $(\ell-1)$th layer and σ is the LeakyReLU activation function. $H_i^{(\ell)} \in \mathbb{R}^{K\times d}$ represents the graph node feature matrix at the ℓth graph convolutional layer for the ith ($i = 1, 2$) time phase. $\tilde{A} = A + I$, where A is the constructed spatial–spectral adjacency matrix and I is the identity matrix. $\hat{D}$ is the degree matrix of $\tilde{A}$.

In the process of contrastive learning, a key is the construction of positive and negative sample pairs. We define pixel pairs in the unchanged regions between bitemporal images as positive sample pairs, and negative sample pairs as pixel pairs corresponding to changed regions. By forcing the feature representations of positive sample pairs closer together and feature representations of negative sample pairs far apart, it can effectively capture essential differences between changed and unchanged regions and provide discriminative feature representations for the CD task.

The output features of the last graph convolutional layer are transformed into pixel-level features using the association matrix. Denote $\mathcal{P}$ and $\mathcal{N}$ as the positive and negative sample sets, respectively, for positive sample pairs $(z_{p_1}, z_{p_2}) \in \mathcal{P}$ and negative sample pairs $(z_{n_1}, z_{n_2}) \in \mathcal{N}$, their cosine similarities are calculated as follows:

$$\begin{cases} s^p = \cos\left(z_{p_1}, z_{p_2}\right) \\ s^n = \cos\left(z_{n_1}, z_{n_2}\right). \end{cases} \tag{11}$$

The loss function is then computed as follows:

$$L_{\text{info}} = -\frac{1}{|\mathcal{P}|}\sum_{i=1}^{|\mathcal{P}|} \log\left(\frac{\exp\left(s_i^p\right)}{\exp\left(s_i^p\right) + \sum_{j=1}^{|\mathcal{N}|}\exp\left(s_j^n\right)}\right) \tag{12}$$

where $|\mathcal{P}|$ and $|\mathcal{N}|$ refer to the number of positive and negative sample pairs, respectively, s_i^p is the similarity of the ith positive sample pair, and s_j^n is the similarity of the jth negative sample pair.

C. DFE Module

By performing the different operation on the bitemporal HSIs, the spectral difference information can be obtained, and this information help the network capture the subtle spectral variations.

First, the different map is obtained by subtracting two HSIs and is denoted by $X_3 = |X_1 - X_2|$ and $X_3 \in \mathbb{R}^{H\times W\times B}$. The shallow feature extracted from the different feature X_3 can be represented as follows:

$$X_3' = \text{DSConv}\left(X_3\right) \tag{13}$$

where DSConv refers to the depthwise separable convolution, which consists of a depthwise convolution followed by a pointwise convolution [43].

The resulting feature $X_3' \in \mathbb{R}^{H\times W\times d}$ is then fed into the DFE module, which consists of channel attention (CA), SA, and residual connections. As shown in Fig. 1, the differential feature X_3' is added with the residual feature enhanced by CA and SA to obtain the enhanced differential feature, which can be denoted by $T \in \mathbb{R}^{H\times W\times d}$

$$T = \text{CA}\left(X_3'\right) \odot X_3' + \text{SA}\left(X_3'\right) \odot X_3' + X_3' \tag{14}$$

where $\odot$ denotes the dot product operation.

By employing CA and SA, we focus on important spectral and spatial features of the differential information, while suppressing the irrelevant redundant information.

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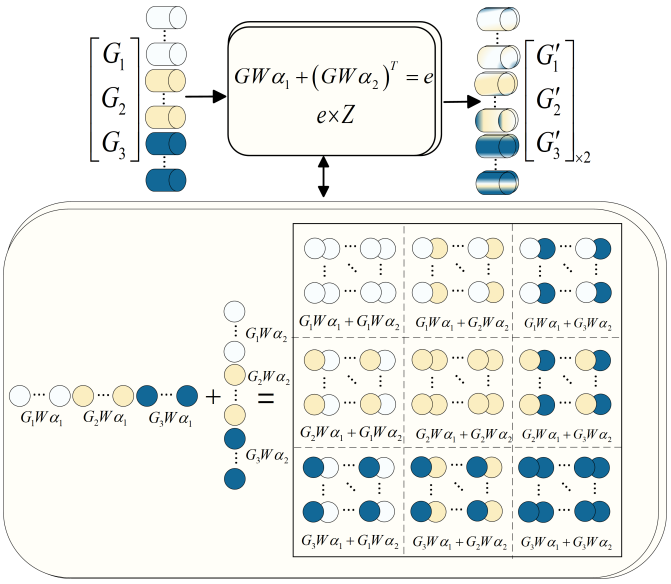


Fig. 2. MGIF module for the interactive fusion of bitemporal graph features with different graph features.

D. MGIF Module

To effectively fuse different features, we employ a graph attention mechanism to adaptively assign information dissemination weights among nodes.

First, the enhanced differential information T needs to be converted into a graph structure utilizing the association matrix Q. The process of obtaining the differential graph can be expressed as follows:

$$G_3 = Q^{\mathrm{T}} T \tag{15}$$

where $G_3 \in \mathbb{R}^{K\times d}$. To enhance graph features and to make full use of different graph information, an MGIF module is used to interact with graph features, as shown in Fig. 2. We denote two graph features obtained from the CFLR module as G_1 and G_2. To facilitate graph features interaction, we concatenate G_1, G_2, and the differential graph G_3 along the sequence dimension to obtain the joint graph node matrix

$$G = [G_1 \parallel G_2 \parallel G_3]. \tag{16}$$

Here, $G \in \mathbb{R}^{3K\times d}$. The sequential dimension refers to the dimension that represents the number of superpixels in the HSIs.

Next, graph features are projected into a shared embedding space using a learnable matrix $W \in R^{d\times d'}$, resulting in $Z = GW \in \mathbb{R}^{3K\times d'}$. d' denotes the dimensionality of the shared embedding space. Two learnable attention vectors $\alpha_1, \alpha_2 \in \mathbb{R}^{d'}$ are introduced to compute the self-attention scores as follows:

$$f_1 = Z\alpha_1, \ \ f_2 = Z\alpha_2. \tag{17}$$

Then, the attention coefficient matrix is calculated as follows:

$$\begin{cases} e = \sigma\left(f_1 + f_2^{\mathrm{T}}\right) \\ \hat{e} = \varphi(e) \end{cases} \tag{18}$$

where $\hat{e}_{i,j} \in \mathbb{R}^{3K\times 3K}$ denotes the normalized attention weight from node S_i to node S_j and φ is the softmax activation function. Based on the attention coefficient matrix $\hat{e}$, the graph node features are then updated as follows:

$$g'_i = \psi\left(\sum_{j=1}^{3K} \hat{e}_{i,j} Z_j\right) \tag{19}$$

where ψ denotes the log-softmax activation function. To further enhance the representation ability, a multihead self-attention mechanism is adopted

$$g''_i = \|_{m=1}^{M} \sigma\left(\sum_{j=1}^{3K} \hat{e}_{i,j}^{(m)} Z_j^{(m)}\right) \tag{20}$$

where M denotes the number of heads and $\|$ denotes concatenating along the channel dimension. The final feature g'' is split to obtain interaction features $G'_1 \in \mathbb{R}^{K\times 2h}$, $G'_2 \in \mathbb{R}^{K\times 2h}$, and $G'_3 \in \mathbb{R}^{K\times 2h}$, where h is the dimension of the heads.

E. Fusion and Loss Function

At the end, the interaction features are projected to pixel-level features and then concatenated with different features, which can be expressed as follows:

$$F = \mathrm{Concat}(P_1, P_2, P_3, T) \tag{21}$$

where Concat$(\cdot)$ refers to the operation of concatenating along the feature dimension and $P_i = QG'_i$.

Then, the fusion feature is fed into a softmax layer to get the predicted labels. The CD is essentially a classification task; thus, the cross-entropy loss function is utilized to penalize the differences between the network output and the true labels of the labeled samples as follows:

$$L_{\mathrm{ce}} = -\frac{1}{n}\sum_{i=1}^{n}\left(y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right) \tag{22}$$

where n denotes the number of labeled samples and y and $\hat{y}$ denote the true label and the predicted label, respectively.

Finally, the contrastive loss L_{info} is combined with the cross-entropy loss L_{ce} to optimize the GCLMA as follows:

$$L = L_{\mathrm{ce}} + \eta L_{\mathrm{info}} \tag{23}$$

where η is a hyperparameter.

III. Experiments

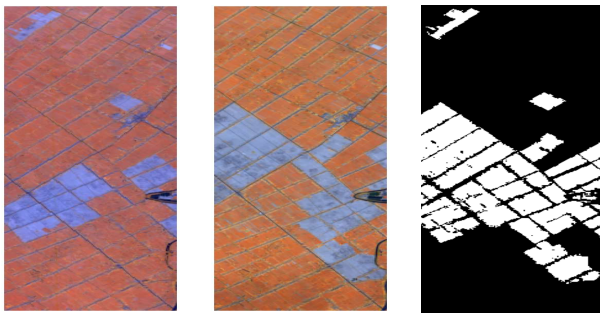
A. Dataset Description

The proposed GCLMA is validated on three public datasets, including the Farmland dataset [44], River dataset [19], and USA dataset [44].

The Farmland dataset was taken in Yancheng, Jiangsu, China. It reflects the changes in farmland between two HSIs, which were obtained on May 3, 2006 and April 23, 2007, respectively. This dataset has a spatial size of 450 × 140 and contains 155 spectral bands, as shown in Fig. 3(a) and (b). The ground truth is shown in Fig. 3(c). It contains 44723 changed pixels and 18277 unchanged pixels.

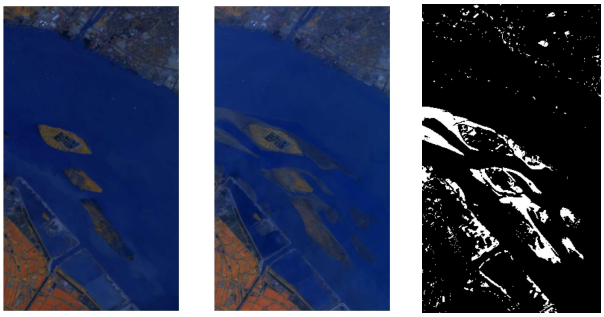
The river dataset was also taken in Jiangsu, China. It reflects changes in the river basin between two HSIs captured on May 3, 2013 and December 31, 2013, respectively. This

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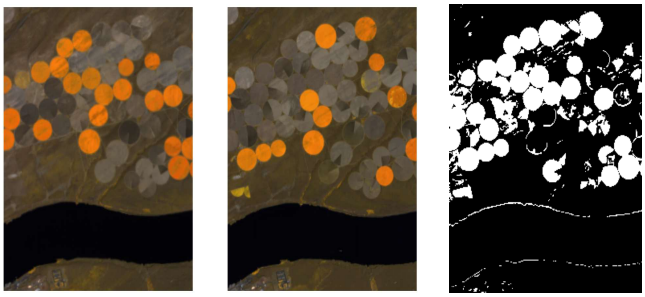
(a) (b) (c)

Fig. 3. Farmland dataset. (a) Image acquired on May 3, 2006. (b) Image acquired on April 23, 2007. (c) Ground-truth image.



(a) (b) (c)

Fig. 4. River dataset. (a) Image acquired on May 3, 2013. (b) Image acquired on December 31, 2013. (c) Ground-truth image.



(a) (b) (c)

Fig. 5. USA dataset. (a) Image acquired on May 1, 2004. (b) Image acquired on May 8, 2007. (c) Ground-truth image.

dataset has a spatial size of 463 × 241 and contains 198 spectral bands. The pseudo-color visualizations can be seen from Fig. 4(a) and (b). There are 9698 changed pixels and 101885 unchanged pixels in the ground truth, as shown in Fig. 4(c).

USA dataset was taken in Hermiston, OR, USA. The HSIs were taken on May 1, 2004 and May 8, 2007, respectively. Both HSIs have a spatial resolution of 307 × 241 pixels with 164 spectral bands. In the ground truth, there are 16676 changed pixels and 57311 unchanged pixels. USA dataset can be seen from Fig. 5.

B. Evaluation Indicators and Experimental Details

1) Evaluation Indicators: Five common metrics are utilized to evaluate the performance, namely, overall accuracy (OA), Kappa coefficient (κ), $F1$, precision, and recall. These metrics can be calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{24}$$

$$\kappa = \frac{\mathrm{OA} - P_e}{1 - P_e} \tag{25}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{26}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{27}$$

$$\mathrm{F1} = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{28}$$

where P_e is computed as follows:

$$P_e = \frac{(\mathrm{TP} \cdot \mathrm{FN}) + (\mathrm{TP} \cdot \mathrm{FP}) + (\mathrm{TN} \cdot \mathrm{FN}) + (\mathrm{TN} \cdot \mathrm{FP})}{N^2}. \tag{29}$$

True positive (TP) refers to the pixels that are correctly identified as "change." False positive (FP) is unchanged pixels incorrectly classified as "change." True negative (TN) is unchanged pixels correctly classified as "no change." False negative (FN) represents the changed pixels that are incorrectly classified as "no change."

2) Experimental Details: The experiments are performed on the NVIDIA GeForce RTX 4060 with a PyTorch framework. The Adam optimizer is used to train the network with a learning rate of 0.0005. The number of training epochs is set to 300. The segmentation scales of the Farmland, River, and USA datasets are set to 15, 30, and 45, respectively. The number of graph convolution layers is 2, and the hidden dimension is consistently set to 128 in CFIR and DFE. The number of heads in MGIF is 2, and the dimension of each head is 32. α is set to 0.8 and η is set to 0.5.

To demonstrate the effectiveness of the proposed method, ten algorithms are used for comparison: CVA [8], MAD [45], ConvLSTM [46], GTMSiam [47], MSDFFN [18], DIENET [48], CSDBF [39], D^2AGCN [36], MGCN [49], and GTransCD [38]. CVA and MAD are traditional methods, while the others are deep learning-based methods. Notably, CSDBF, D^2AGCN, MGCN, and GTransCD are graph neural networks (GNNs)-based methods.

C. CD Performance

We randomly select 1% labeled pixels for training. Optimal results in the table are shown in boldface, and suboptimal results are underlined. Red pixels in the visualized result images indicate FNs and blue pixels indicate FPs.

1) Farmland Dataset: The quantitative and qualitative comparisons on the Farmland dataset are presented in Table I and Fig. 6, respectively. The experimental results show that traditional algorithms (e.g., CVA and MAD) exhibit satisfactory performance. However, these methods still lag behind deep learning-based methods, especially in terms of precision and $F1$. It is worth noting that GTransCD achieves the highest recall, indicating its stronger ability to detect changed regions. However, its relatively lower precision suggests a higher rate of FPs. In general, our GCLMA has the most outstanding

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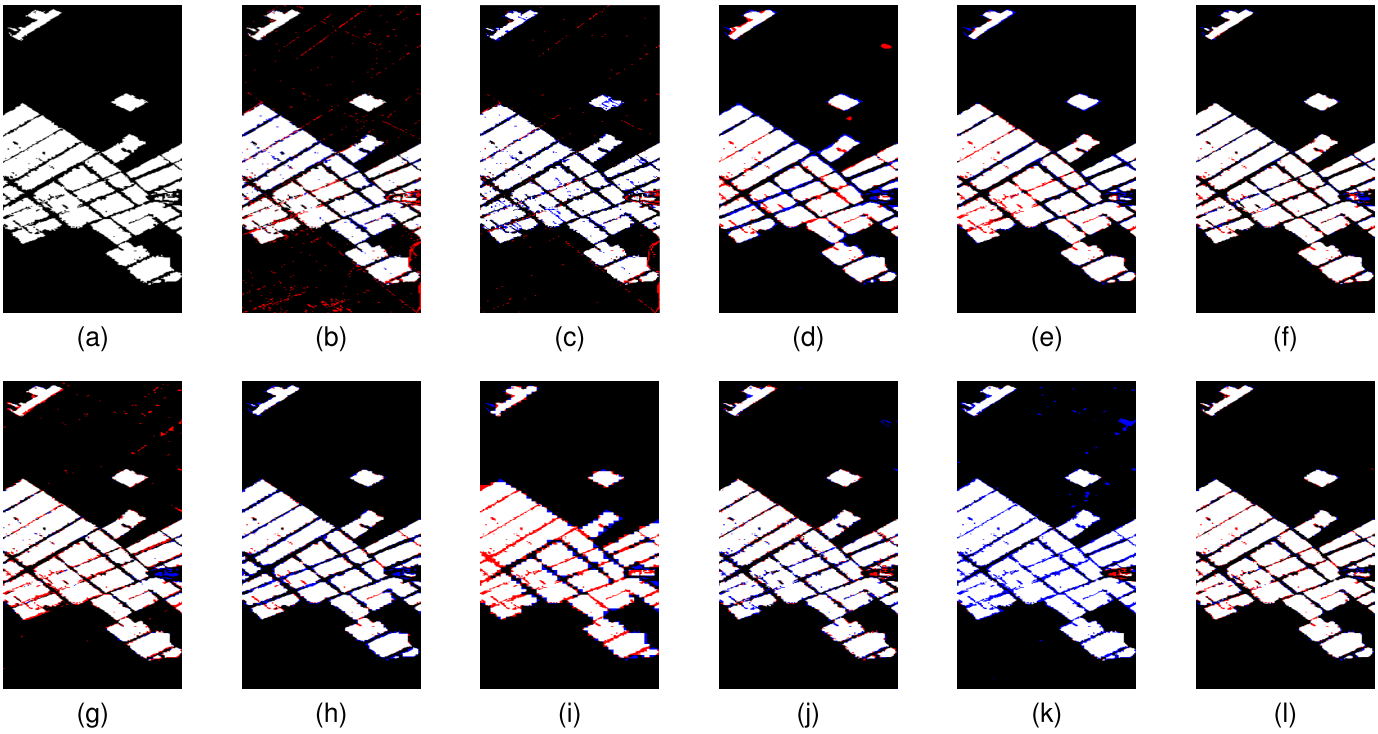


Fig. 6. CD maps of all methods on the Farmland dataset with 1% labeled samples. (a) Ground truth. (b) CVA. (c) MAD. (d) ConvLSTM. (e) GTMSiam. (f) MSDFFN. (g) DIEFEN. (h) CSDBF. (i) D^2AGCN. (j) MGCN. (k) GTransCD. (l) GCLMA. The FN and FP pixels are denoted by red and blue, respectively.

TABLE I
CD RESULTS ON THE FARMLAND DATASETS WITH 1% LABELED SAMPLES

Methods	OA (%)	κ ($\times$100)	F1 (%)	Precision (%)	Recall (%)
CVA	95.74	89.84	92.87	90.36	95.51
MAD	96.35	91.04	93.59	95.44	91.80
ConvLSTM	95.60±0.47	89.28±0.01	92.37±0.83	92.82±1.23	91.96±1.57
GTMSiam	96.53±0.10	91.65±0.20	94.11±0.12	92.60±0.81	95.68±0.70
MSDFFN	96.51±0.19	91.64±0.44	94.12±0.30	91.95±0.95	96.42±0.79
DIEFEN	96.61±0.24	91.92±0.56	94.34±0.39	91.44±0.63	97.42±0.27
CSDBF	96.55±0.11	91.61±0.25	94.36±0.57	93.49±1.44	94.33±1.25
D^{2}AGCN	93.55±0.12	84.82±0.23	95.36±0.10	97.49±0.20	93.32±0.37
MGCN	95.54±0.15	88.35±0.26	91.99±0.16	89.69±1.72	94.48±1.83
GTransCD	94.74±0.45	87.06±0.95	91.06±0.62	84.59±1.48	**98.63±0.88**
GCLMA	**97.18±0.08**	**93.23±0.20**	**98.00±0.06**	**98.85±0.19**	97.16±0.19

overall performance on the Farmland dataset, especially in terms of OA and precision. This is mainly attributed to the CFLR and MGIF modules, where the former helps to capture the change information and the latter fuses multigraph features interactively to enrich the feature representation.

Fig. 6 shows that GCLMA achieves the best visual result among all methods. Traditional methods such as CVA and MAD produce high FPs and FNs. Notably, CVA, DIEFEN, and D^2AGCN have many FN pixels, such as the edge of the farmland. Moreover, CVA, ConvLSTM, and DIEFEN tend to misclassify unchanged pixels as changed ones (FPs), especially in background areas. While CSDBF, MGCN, and MSDFFN show improved performance, they still produce some spurious detections in the small farmland region on the right side of the image. Overall, GCLMA demonstrates the fewest FPs and FNs, resulting in the most accurate and visually coherent CD map.

2) River Dataset: Table II shows the quantitative results on the River dataset. Traditional algorithms perform well in terms of OA, but show poor results in terms of κ, $F1$, and recall. This may be attributed to their reliance on thresholds, which are different to determine. In contrast, deep learning-based methods, such as GTMSiam, DIEFEN, and CSDBF, demonstrate relatively better performance on both OA and κ. Among them, DIEFEN achieves an outstanding result in recall, indicating its superior ability to detect changed regions. All methods show relatively low κ, indicating limited agreement between predictions and ground truth. Our GCLMA

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TABLE II
CD RESULTS ON THE RIVER DATASETS WITH 1% LABELED SAMPLES

Methods	OA (%)	κ ($\times$100)	F1 (%)	Precision (%)	Recall (%)
CVA	95.52	66.08	68.36	88.42	55.72
MAD	96.05	71.53	73.60	87.89	63.31
ConvLSTM	95.06±0.19	64.91±0.02	67.51±1.77	78.83±2.46	59.16±3.16
GTMSiam	95.93±0.05	71.72±0.87	73.89±0.87	83.58±1.35	66.27±2.19
MSDFFN	95.31±0.20	65.81±1.51	68.26±1.43	83.02±2.07	57.98±1.89
DIEFEN	96.15±0.24	75.66±0.94	77.77±0.82	78.22±3.15	77.46±2.09
CSDBF	96.11±0.20	73.92±1.17	76.03±1.07	81.95±2.60	70.96±1.61
D^2AGCN	93.78±0.05	55.55±0.35	58.84±0.32	69.29±0.47	51.12±0.28
MGCN	94.61±0.31	64.85±2.14	67.82±2.00	68.29±2.04	67.45±3.35
GTransCD	95.99±0.27	72.98±2.53	75.15±2.40	81.27±0.64	70.00±4.52
GCLMA	**96.96±0.17**	**80.42±1.30**	**82.08±1.22**	**84.22±1.27**	**80.09±2.49**

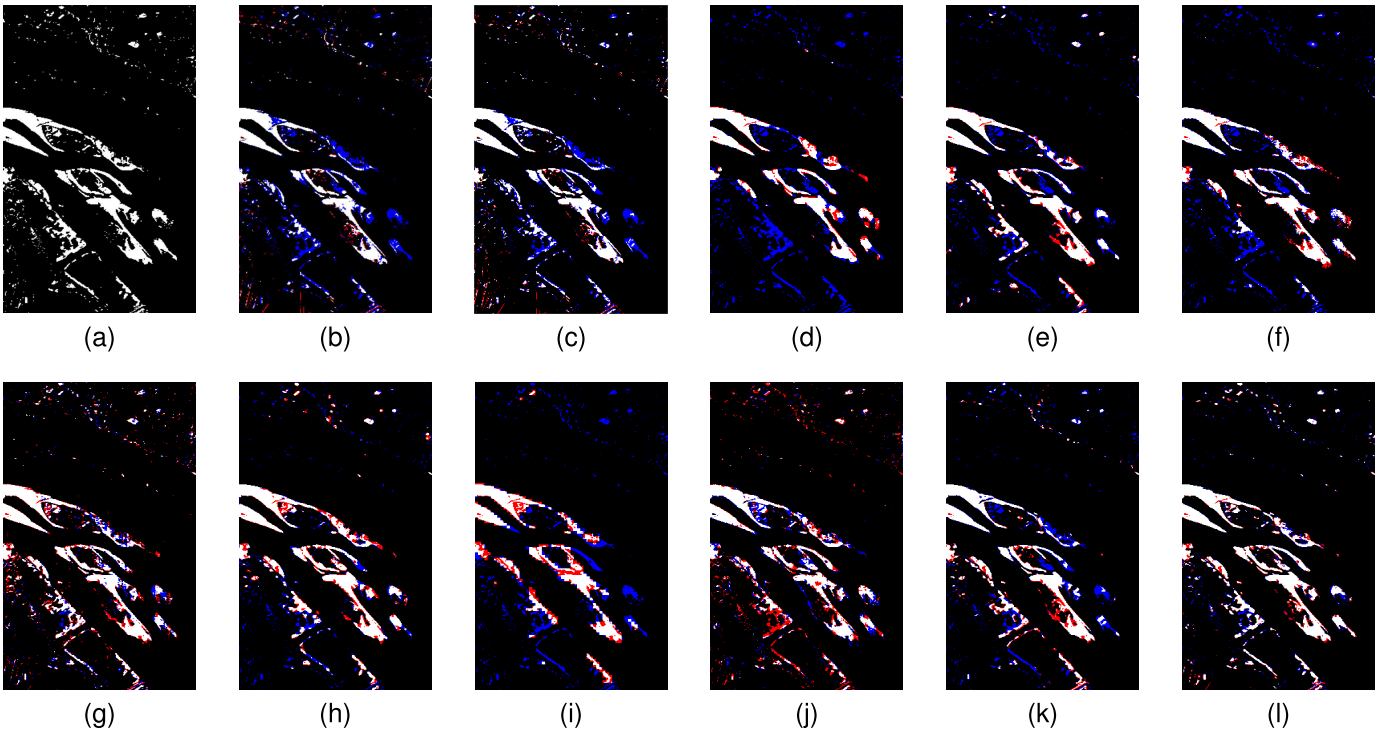


Fig. 7. CD maps of all methods on the River dataset with 1% labeled samples. (a) Ground truth. (b) CVA. (c) MAD. (d) ConvLSTM. (e) GTMSiam. (f) MSDFFN. (g) DIEFEN. (h) CSDBF. (i) D^2AGCN. (j) MGCN. (k) GTransCD. (l) GCLMA. The FN and FP pixels are denoted by red and blue, respectively.

achieves the highest performance across multiple key metrics. Specifically, GCLMA achieves κ score exceeding 80%, representing an improvement of approximately 4.76% over the second-best method. Moreover, the GCLMA achieved the highest $F1$, indicating an excellent balance between precision and recall. This reflects its capability to suppress false alarms while maintaining sensitivity to changes. In general, GCLMA performs the best on the River dataset.

Fig. 7 presents the CD maps on the River dataset. As shown in Fig. 7, DIEFEN, CSDBF, and GCLMA are more effective at identifying changed regions. However, DIEFEN, CSDBF, and MGCN have more FN pixels along the boundaries of exposed land, reflecting their weaker capability in edge detection. In contrast, our GCLMA yields the fewest FN and FP pixels (highlighted in red and blue).

3) USA Dataset: Table III presents the results on USA dataset. It can be observed that the traditional algorithms, i.e., CVA and MAD, show good performance in recall but average performance in OA. In contrast, the deep learning-based methods are more capable of distinguishing between the positive and negative samples, resulting in both higher recall and OA. CSDBF significantly outperforms D^2AGCN. MGCN and GTransCD achieve comparable OA, but GTransCD shows slightly higher κ and $F1$ than MGCN. GTMSiam achieves high precision but relatively low recall. Although the overall performance of GTransCD is comparable, its $F1$ and κ are

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TABLE III
CD RESULTS ON USA DATASETS WITH 1% LABELED SAMPLES

Methods	OA (%)	κ ($\times$100)	F1 (%)	Precision (%)	Recall (%)
CVA	92.70	81.19	86.01	75.70	99.58
MAD	91.62	78.75	84.28	72.97	99.75
ConvLSTM	93.66±0.29	81.23±1.08	85.25±0.92	89.64±1.10	81.33±2.39
GTMSiam	95.53±0.12	86.81±0.31	89.65±0.25	**93.85±1.62**	85.84±1.47
MSDFFN	94.68±0.21	84.66±0.49	88.08±0.35	89.10±1.70	87.12±1.30
DIEFEN	95.08±0.21	86.02±0.56	89.20±0.44	88.39±1.40	90.06±1.50
CSDBF	95.33±0.25	86.38±0.73	89.38±0.58	91.61±1.53	87.28±1.62
D^2AGCN	91.43±0.17	74.34±0.54	79.75±0.45	85.31±0.73	74.88±0.97
MGCN	95.68±0.15	83.56±0.47	87.30±0.39	92.34±1.17	82.81±1.33
GTransCD	95.46±0.43	86.73±1.12	89.64±0.84	92.34±2.18	87.12±0.72
GCLMA	**96.67±0.17**	**90.41±0.42**	**92.55±0.31**	93.24±1.37	**91.90±0.95**

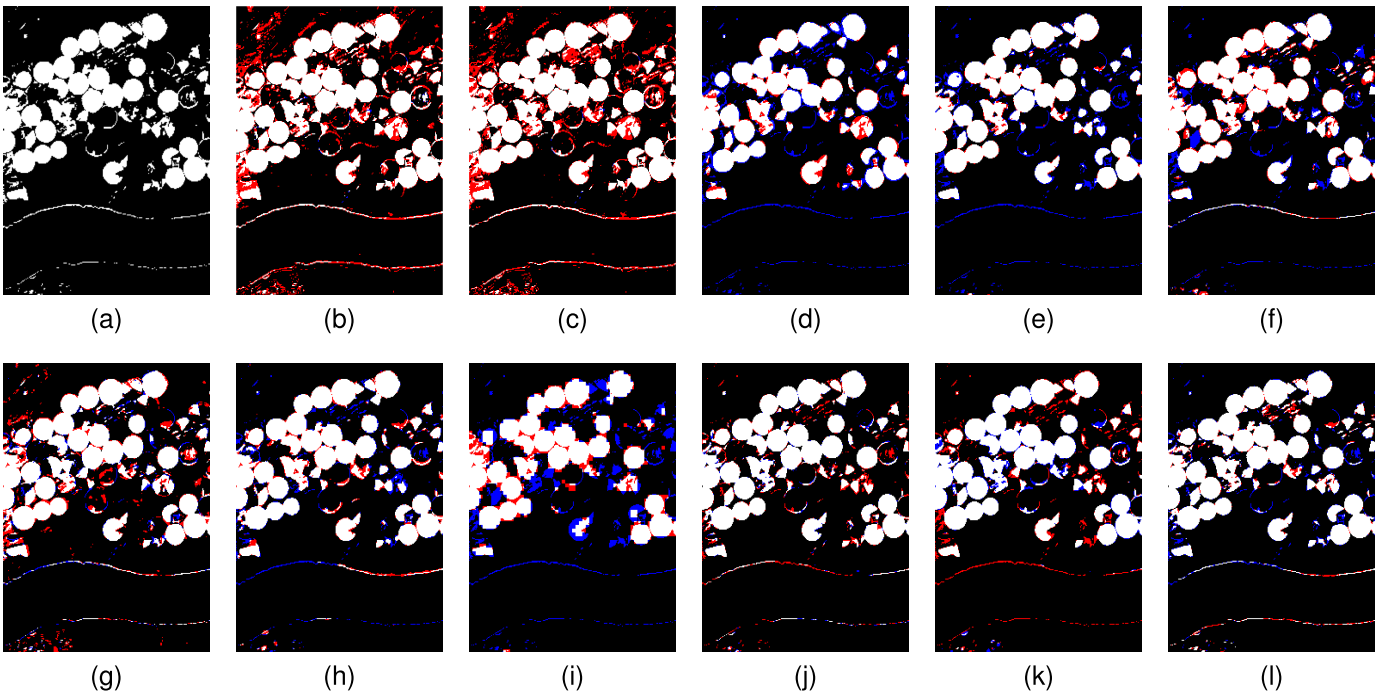


Fig. 8. CD maps of all methods on USA dataset with 1% labeled samples. (a) Ground truth. (b) CVA. (c) MAD. (d) ConvLSTM. (e) GTMSiam. (f) MSDFFN. (g) DIEFEN. (h) CSDBF. (i) D^2AGCN. (j) MGCN. (k) GTransCD. (l) GCLMA. The FN and FP pixels are denoted by red and blue, respectively.

lower than those of GCLMA. Overall, GCLMA shows the best overall performance.

The visual results in Fig. 8 demonstrate that traditional algorithms struggle with a high-false detection rate, as evidenced by the large number of red areas shown in Fig. 8(b) and (c). In general, deep learning algorithms outperform traditional algorithms. For example, ConvLSTM, MSDFFN, GTMSiam, MGCN, and GTransCD have fewer FN pixels. However, they also produce more FP pixels. In contrast, although GCLMA has slightly more FN pixels, it produces significantly fewer FP pixels, resulting in the best visual result.

D. Ablation Studies and Analysis

1) Effective of Key Modules: Ablation experiments are conducted to demonstrate the effectiveness of the proposed modules, and the results are shown in Table IV. After using the SSGC module, OA exceeds 96% on the Farmland and River datasets and 95.5% on USA. The combination of SSGC and CFLR shows that the use of contrastive learning can help the model highlight change information and thus improve the performance. The combination of SSGC and MGIF shows that interacting with different features with the bitemporal features can help the network to learn richer features. In addition, our results show that interacting features via graph attention yields better performance. By integrating three modules, the proposed GCLMA achieves the best performance across the three datasets.

2) Results of Different Sampling Rates: To further validate the robustness of GCLMA, we conduct experiments under different sampling rates. Table V shows the values of OA corresponding to 0.5%, 1%, 2%, and 5% sampling rates for

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TABLE IV
ABLATION STUDY RESULTS ON DIFFERENT DATASETS

Dataset	SSGC	CFLR	MGIF	OA (%)	κ ($\times$100)	F1 (%)	Precision (%)	Recall (%)
Farmland	✓	×	×	96.23±0.13	91.02±0.30	97.31±0.10	98.66±0.19	96.00±0.28
	✓	✓	×	96.61±0.12	91.86±0.29	97.59±0.09	98.34±0.17	96.85±0.25
	✓	×	✓	96.79±0.29	92.31±0.65	97.71±0.21	98.72±0.29	96.73±0.63
	✓	✓	✓	**97.18±0.08**	**93.23±0.20**	**98.00±0.06**	**98.85±0.19**	**97.16±0.19**
River	✓	×	×	96.31±0.06	75.81±0.39	77.82±0.36	81.40±0.63	74.54±0.57
	✓	✓	×	96.54±0.14	77.03±0.84	78.91±0.78	83.85±2.39	74.59±2.32
	✓	×	✓	96.63±0.06	79.02±0.12	80.87±0.09	79.75±1.26	**82.05±1.20**
	✓	✓	✓	**96.96±0.17**	**80.42±1.30**	**82.08±1.22**	**84.22±1.27**	80.09±2.49
USA	✓	×	×	95.66±0.10	87.48±0.20	90.27±0.14	91.28±1.71	89.32±1.58
	✓	✓	×	96.04±0.27	88.51±0.81	91.04±0.64	92.89±0.33	89.28±1.05
	✓	×	✓	96.32±0.17	89.38±0.59	91.75±0.49	92.58±0.73	90.96±1.59
	✓	✓	✓	**96.67±0.17**	**90.41±0.42**	**92.55±0.31**	**93.24±1.37**	**91.90±0.95**

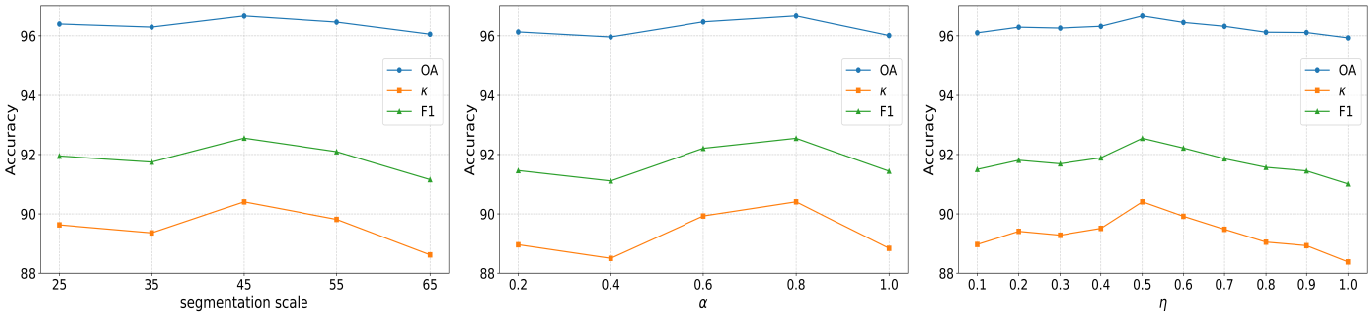


Fig. 9. CD Results for different hyperparameters on USA dataset. (a) Performance of different segmentation scales. (b) Performance of different values of α. (c) Performance of different values of η.

TABLE V
COMPARISON OF OA ACROSS ALGORITHMS WITH DIFFERENT SAMPLING RATES ON 1%USA DATASET

Methods	0.5%	1%	2%	5%
ConvLSTM	92.86	93.66	94.67	95.38
GTMSiam	95.39	95.53	95.99	96.39
MSDFFN	93.40	94.68	95.65	96.60
DIEFEN	93.27	94.75	95.52	96.66
CSDBF	94.36	95.33	95.74	97.09
D^2AGCN	91.24	91.43	92.08	92.87
MGCN	94.43	95.54	96.20	96.81
GTransCD	94.91	95.46	96.20	96.70
GCLMA	**96.30**	**96.67**	**97.20**	**97.62**

all algorithms on USA dataset. As the training sampling rate increases, OA achieved by each algorithm also increases. It is worth noting that GCLMA is always optimal under all training sampling rates.

3) Discussion of Different Segmentation Scales: A large segmentation scale will reduce the computational complexity, but may lose detailed information. In contrast, a small segmentation scale can help to preserve details but increases computational cost. Fig. 9(a) shows the effect of different segmentation scales on USA dataset. It can be seen that the optimal segmentation scale is 45. Correspondingly, the optimal segmentation scales are 15 and 30 for the Farmland dataset and the River dataset, respectively.

4) Discussion of α: α is used to control the fusion ratio of spatial and spectral information in constructing the spatial–spectral graph. Fig. 9(b) shows the effect of different α. As α increases, the spectral information are gradually weakened in the graph construction. The model performs the best when α is set to 0.8. This indicates that proper consideration of spectral information can help to build a well-defined graph structure and enhance the model performance.

5) Discussion of η: η adjusts the contribution of the contrastive loss to the overall objective. Its effect on performance is shown in Fig. 9(c). As η increases from 0.1 to 0.5, there is a consistent improvement in all evaluation metrics. This indicates that appropriately incorporating contrastive learning enhances feature discriminability and contributes positively to the CD performance. The best results are achieved when η equals 0.5. When η exceeds 0.5, the performance slightly

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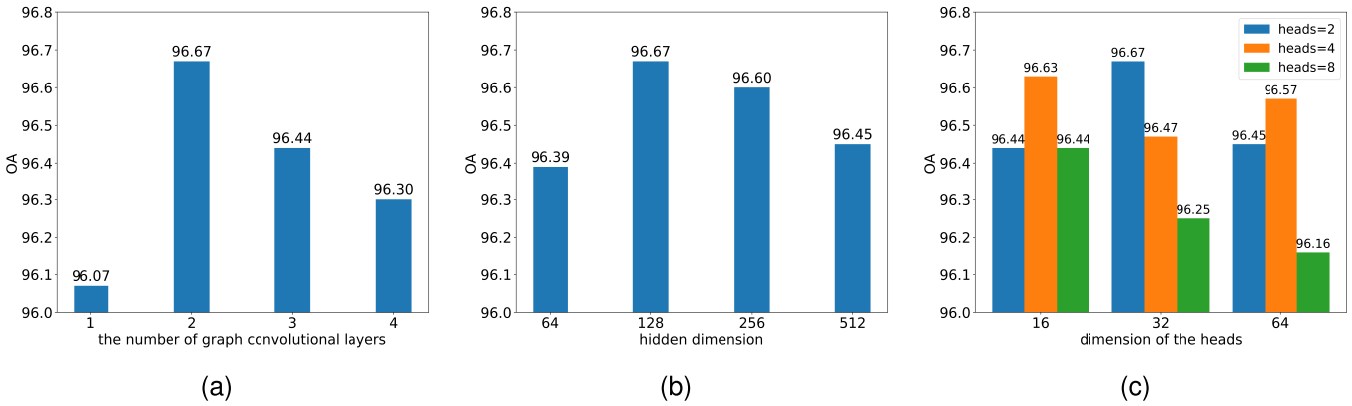


Fig. 10. CD Results for different network hyperparameters on USA dataset. (a) Impact of the number of graph convolutional layers on OA. (b) Impact of hidden dimension on OA in the CFLR and DFE modules. (c) Effect of the number of heads and the dimension of the heads on OA.

TABLE VI
NUMBER OF PARAMETERS AND RUNNING TIME OF DIFFERENT ALGORITHMS ON USA DATASET

Methods	Parameters	Training time	Testing time
ConvLSTM	229,833	1014.39	18.33
GTMSiam	60,532	1647.65	14.50
MSDFFN	26,458,392	993.02	39.24
DIEFEN	8,077,508	4987.46	22.46
CSDBF	105,246	521.62	0.63
D^2AGCN	2,148,303	2049.22	3.42
MGCN	94,714	3181.93	3.61
GTransCD	363,799,691	22623.62	0.03
GCLMA	236,014	46.65	0.01

degrades. This may be because the contrastive learning loss dominates the training process, weakening the effect of supervised learning and thus slightly reducing performance.

6) Discussion of Network Hyperparameters: The network hyperparameters include the number of graph convolutional layers, the hidden dimension, the number of heads, and the dimension of the heads. The results of the effect of different hyperparameters are shown in Fig. 10. The number of graph convolutional layers is set to 2, which can balance model complexity and performance. Based on experimental results, the hidden dimension is set to 128, the number of heads is set to 2, and the dimension of the heads is set to 32.

7) Discussion of Computational Cost: To verify the efficiency of GCLMA, we compare the number of parameters as well as the running time of all algorithms, shown in Table VI. GTransCD and MSDFFN have significantly large number of parameters. GTransCD, DIEFEN, and MGCN need more training time. In summary, our GCLMA shows relatively lower computational cost to achieve much better performance.

IV. CONCLUSION

In this article, we have proposed a GCLMA for HSI-CD. GCLMA first utilizes the SSGC to construct spatial–spectral graphs of bitemporal HSIs and then uses CFLR to enhance the discrimination of features by incorporating contrastive learning and residual structure. Finally, MGIF is used to fully integrate the enhanced difference features and the features of bitemporal HSIs to improve CD performance. Experimental results on three real datasets have demonstrated the effectiveness of the proposed GCLMA.

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